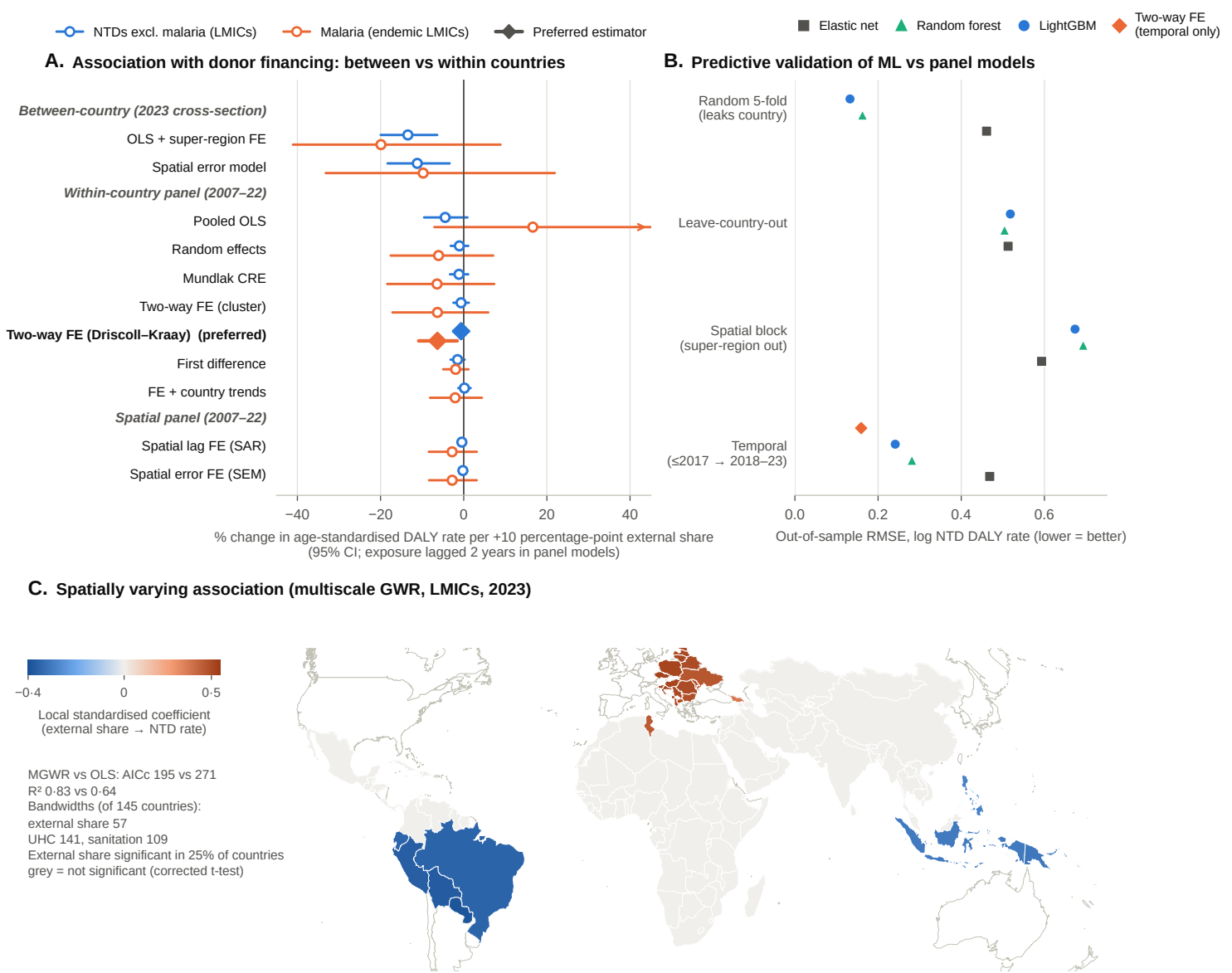

NTDs excl. malaria (LMICs)
Malaria (endemic LMICs)
Preferred estimator
Elastic net
Random forest
LightGBM
Two-way FE (temporal only)
A. Association with donor financing: between vs within countries
Between-country (2023 cross-section)
OLS + super-region FE
Spatial error model
Within-country panel (2007–22)
Pooled OLS
Random effects
Mundlak CRE
Two-way FE (cluster)
Two-way FE (Driscoll–Kraay) (preferred)
First difference
FE + country trends
Spatial panel (2007–22)
Spatial lag FE (SAR)
Spatial error FE (SEM)
−40
−20
0
20
40
% change in age-standardised DALY rate per +10 percentage-point external share
(95% CI; exposure lagged 2 years in panel models)
B. Predictive validation of ML vs panel models
Random 5-fold
(leaks country)
Leave-country-out
Spatial block
(super-region out)
Temporal
(≤2017 → 2018–23)
0.0
0.2
0.4
0.6
Out-of-sample RMSE, log NTD DALY rate (lower = better)
C. Spatially varying association (multiscale GWR, LMICs, 2023)
−0·4
0
0·5
Local standardised coefficient
(external share → NTD rate)
MGWR vs OLS: AICc 195 vs 271
R² 0·83 vs 0·64
Bandwidths (of 145 countries):
external share 57
UHC 141, sanitation 109
External share significant in 25% of countries
grey = not significant (corrected t-test)